

Cola intake and serum lipids in the Oslo Health Study.

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Soft drinks can be a major source of sucrose, which may influence serum lipid concentration. We have examined the association between intake frequency of various types of soft drinks and the concentration of serum triglycerides (TG) and high-density lipoprotein (HDL) and low-density lipoprotein (LDL) cholesterol in the cross-sectional Oslo Health Study. A total of 14 188 subjects of the altogether 18,770 participants of the study had data on intake frequency of colas and non-colas, with or without sugar. The population sample consisted of both sexes and 3 age groups: group 1 (30 years of age), group 2 (40 and 45 years of age), and group 3 (59-60 years of age). In both sexes, HDL decreased and TG increased significantly ($p < 0.001$) with increasing intake frequency of colas. In contrast, no consistent associations were found between the reported intake of non-cola soft drinks and the serum lipids. We found no significant differences related to the reported presence or absence of sugar in the soft drinks. In multiple linear regression analyses, the colas vs. serum lipid associations prevailed ($p < 0.001$) after including 13 possible confounders: sex; age group; time since last meal; physical activity; intake of alcohol, coffee, cheese, fruit and (or) berries, and fatty fish; smoking; length of education; use of cholesterol-lowering drugs; and intake of non-colas. Thus, the self-reported intake frequency of colas, but not other soft drinks, was negatively associated with serum HDL, and positively associated with TG and LDL.